

Data Science

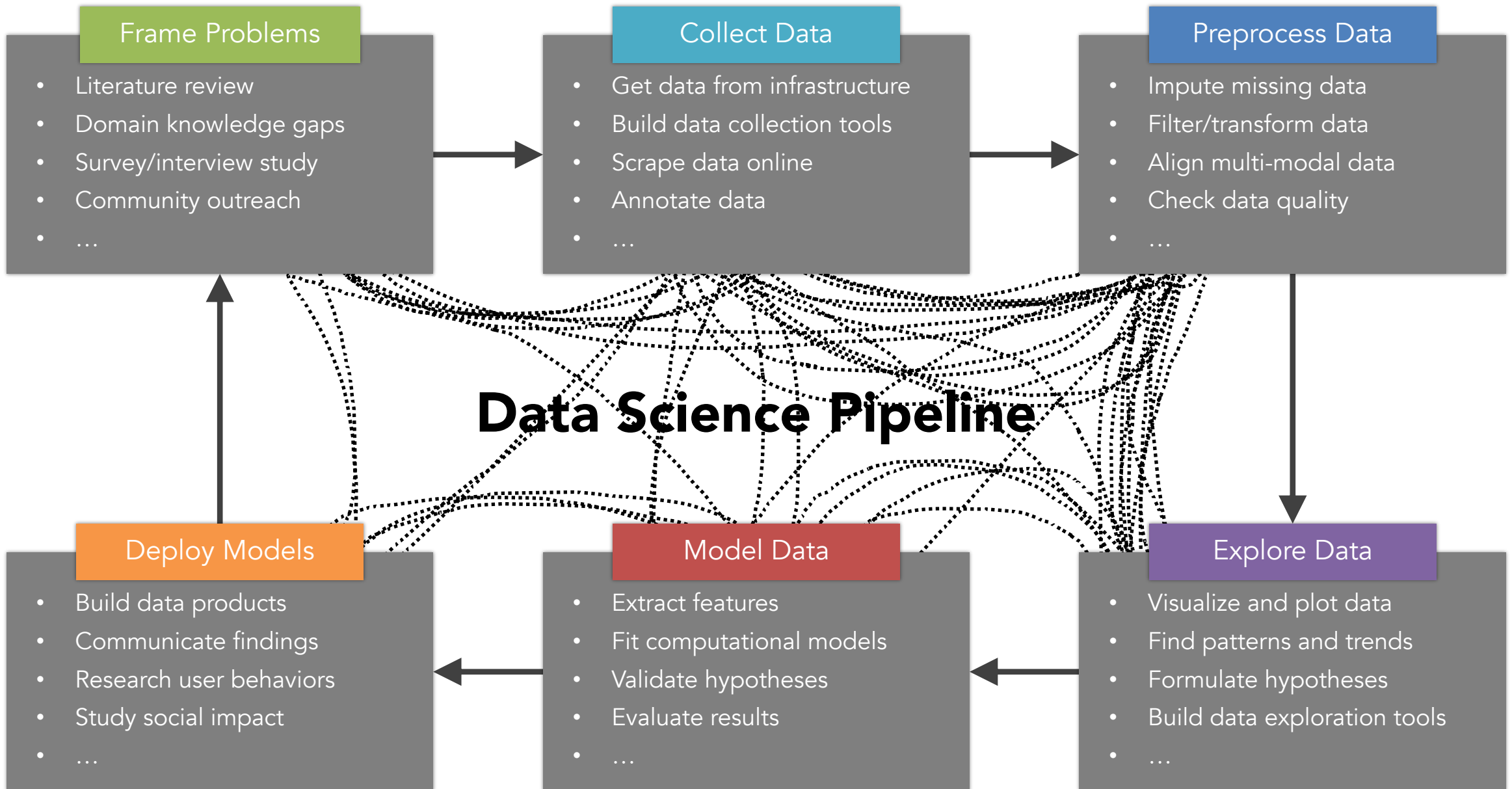
Lecture 1-1: Course Introduction and Organization



Lecturer: Yen-Chia Hsu

Date: Sep 2026

Data science is about turning rich data into actionable insight and making data impactful!



What people typically think : →

The reality of the data science pipeline:

Frame Problems

This course uses existing scenarios and cases with well-defined problems. However, in the real world, we need to define and frame the problems first.



Example: understanding parking patterns and law enforcement -- <https://responsiblesensinglab.org/projects/scan-cars>

 LEGEND
  ANALYSIS

Tree cover gain - 2000-2020

Tree cover gain

Tree cover loss - 2001-2021

Tree cover loss

Displaying Tree cover loss with canopy density > 30%

2001 2004 2008 2011 2015 2018 2021

Tree cover loss is not always deforestation.

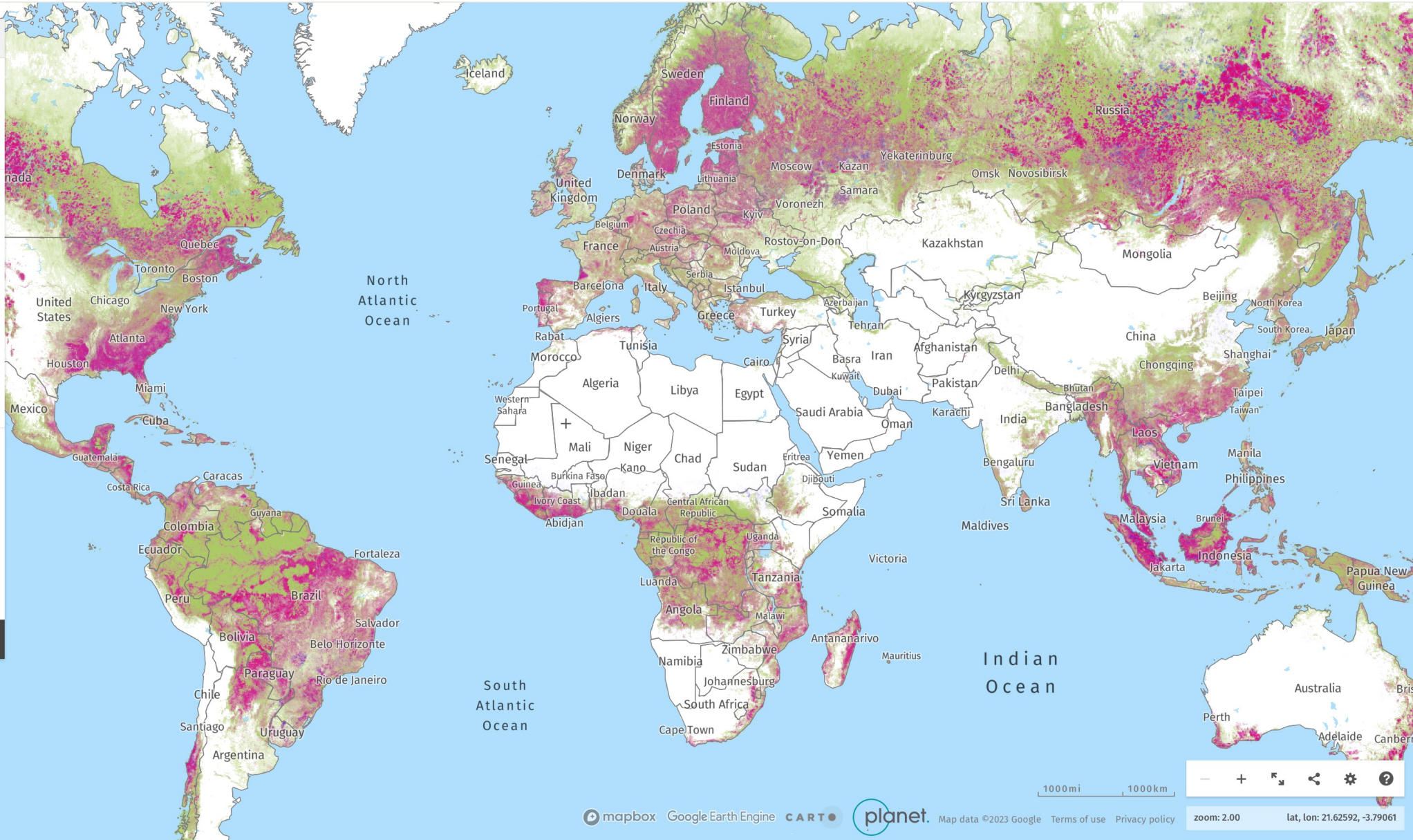
Tree cover - 2010

Tree cover

Displaying Tree cover with canopy density > 30%

Displaying Tree cover for 2010

PLANET SATELLITE IMAGERY (TROPICS)

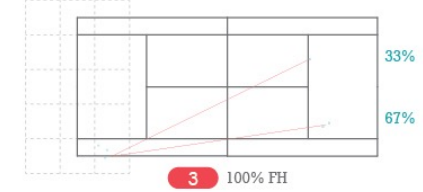
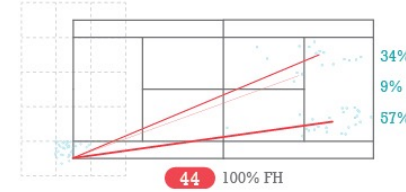
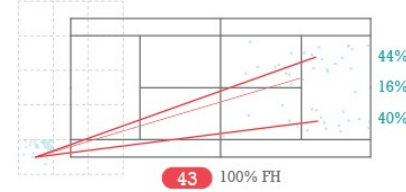
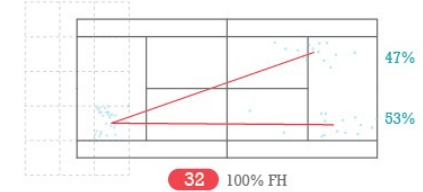
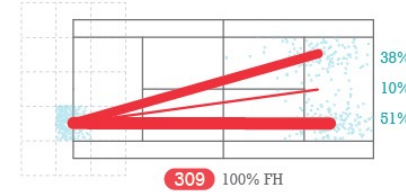
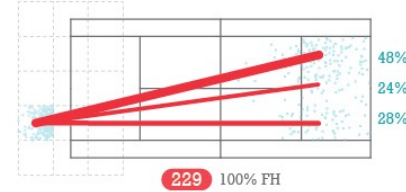
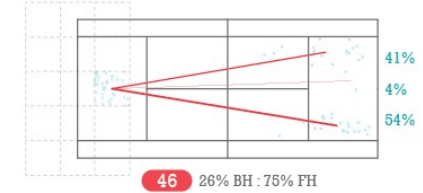
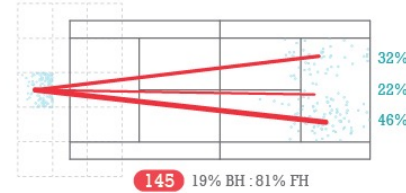
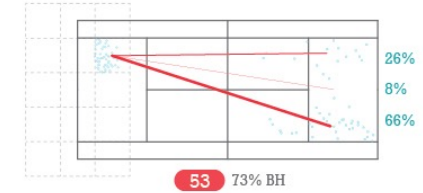
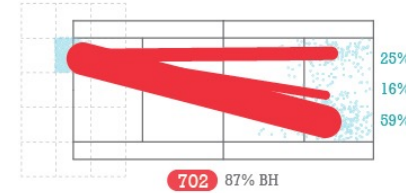
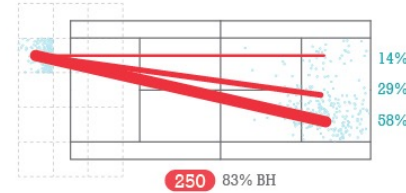
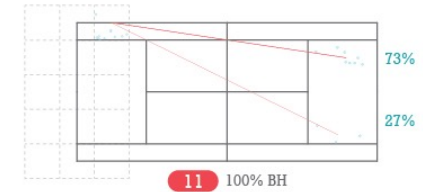
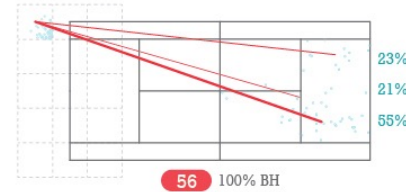
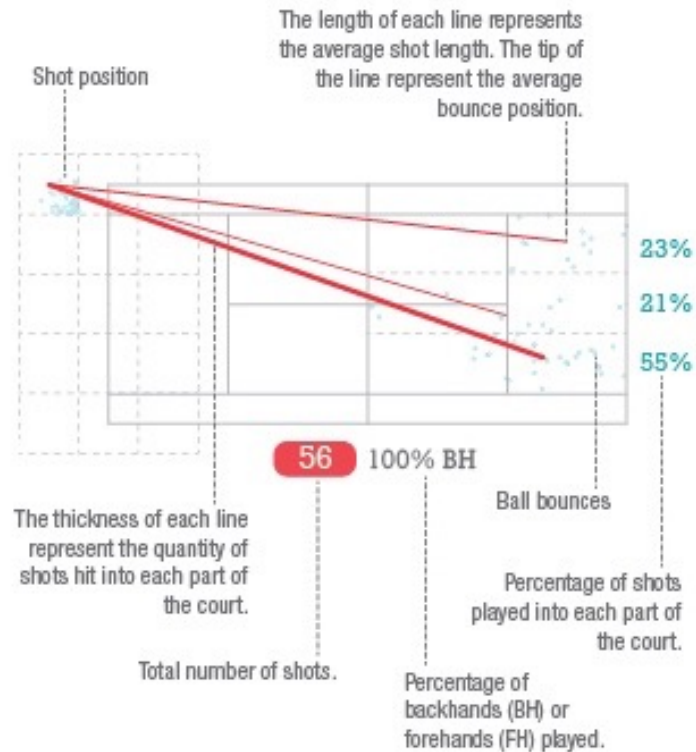




Kei Nishikori Shot Charts

錦織圭のショットチャート

Shot charts are critical in understanding a player's on court behaviour. They are frequently used to map shot patterns from particular areas of the court. These patterns are of particular interest to coaches and players for pre and post match tactical analysis. Here we present 2,443* shots from Kei Nishikori that were played over a period of 6 months in 2014-15 against opponents like Federer, Djokovic, Murray and Wawrinka.

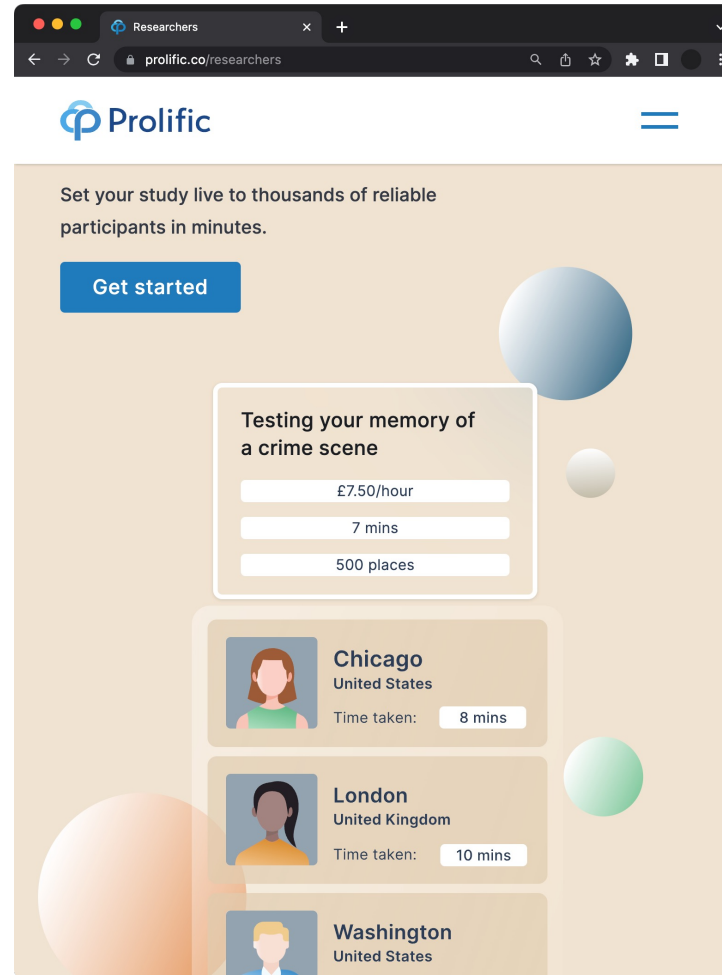


Collect Data

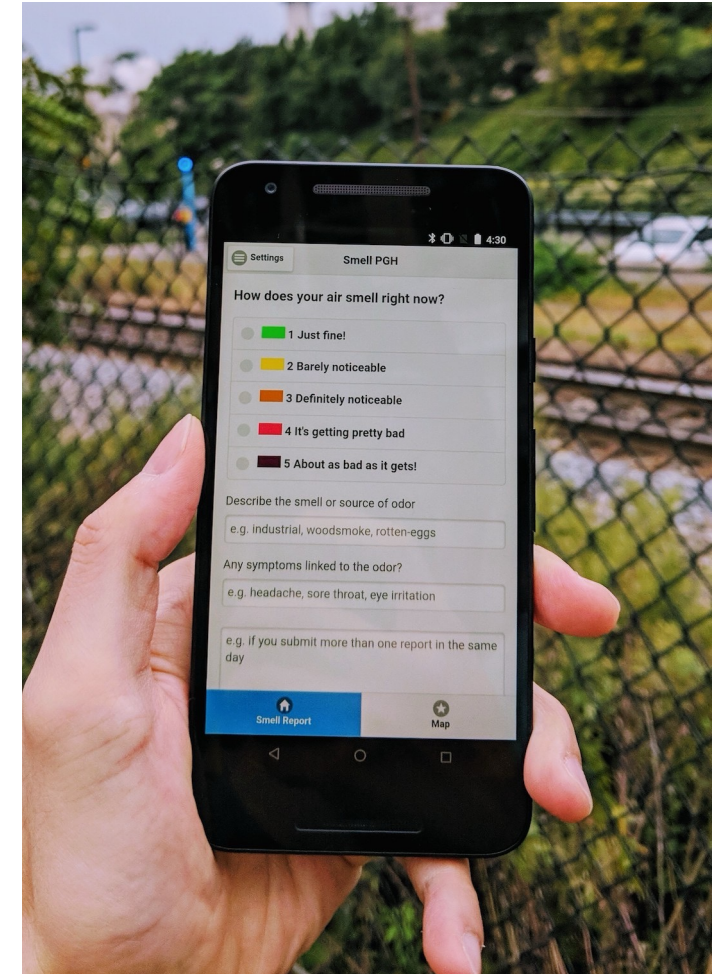
This course assumes that someone has collected the data for you. In reality, you may need to collect data using sensors, crowdsourcing, mobile apps, etc.



[GGD Amsterdam Data Portal](https://data.amsterdam.nl/)



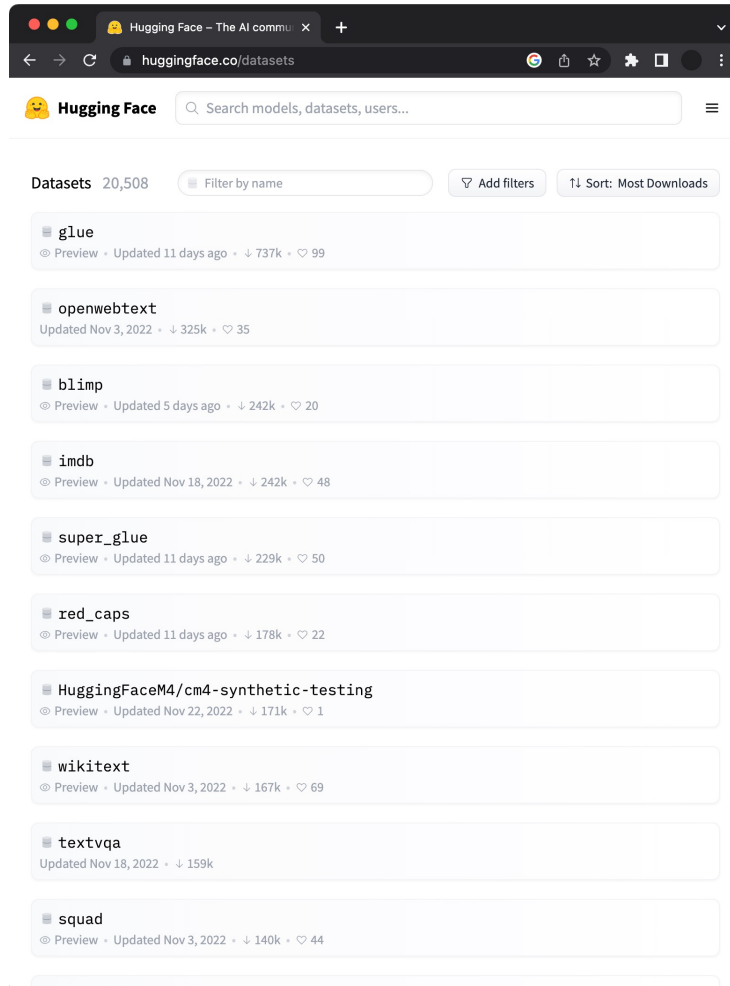
[Prolific Tool for Data Annotation](https://prolific.co/researchers)



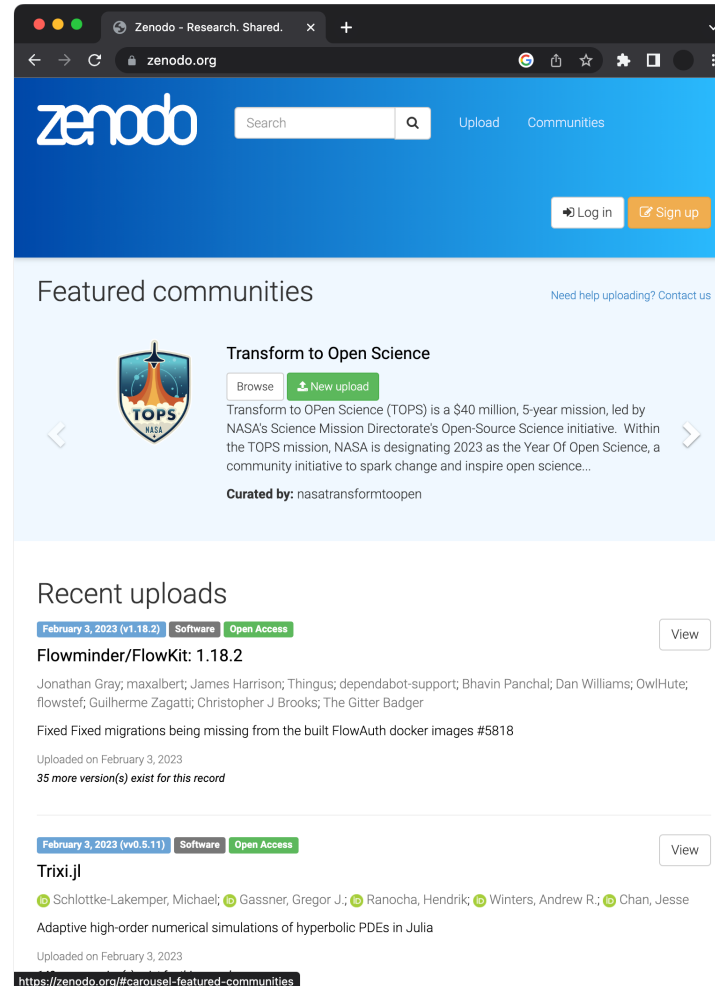
[Mobile App Data Collection](#)

Collect Data

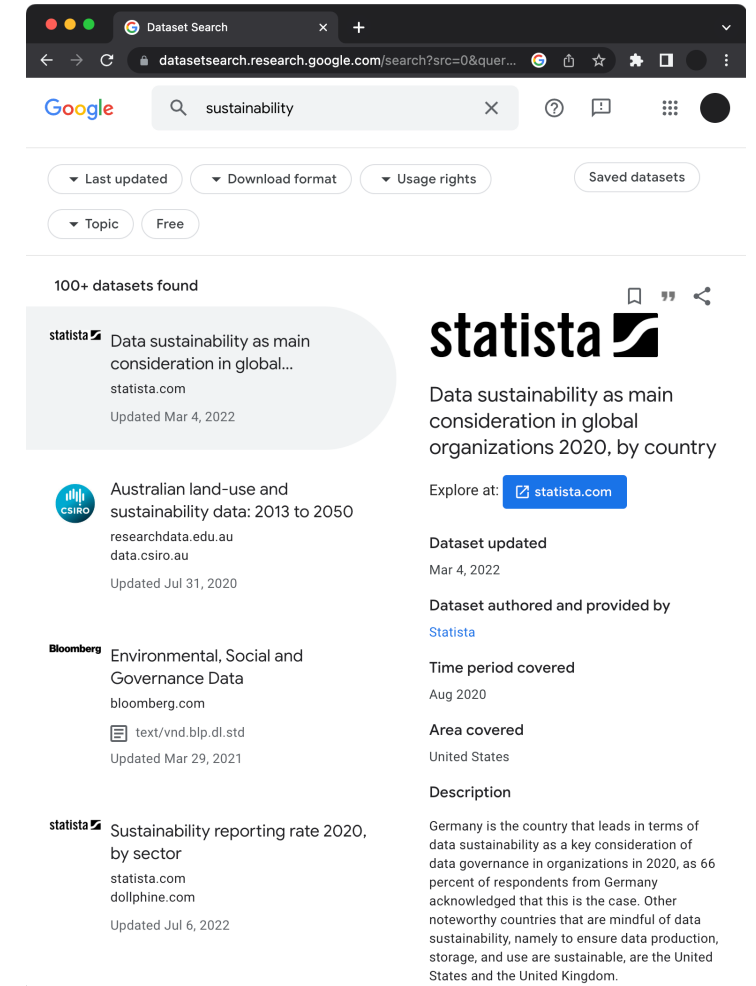
There are also other sources for getting public datasets, such as Hugging Face, Zenodo, Google Dataset Search, government websites, etc.



[Hugging Face](https://huggingface.co/datasets)



[Zenodo](https://zenodo.org)

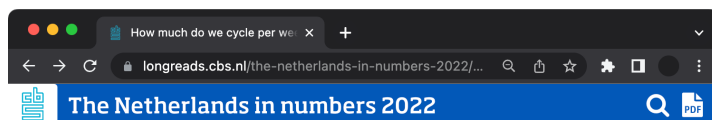


[Google Dataset Search](https://datasetsearch.research.google.com/search?src=0&quer...)

This course will use [pandas](#), which is a very handy Python library for preprocessing structured data. We will cover the following techniques:

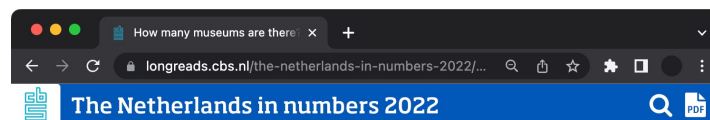
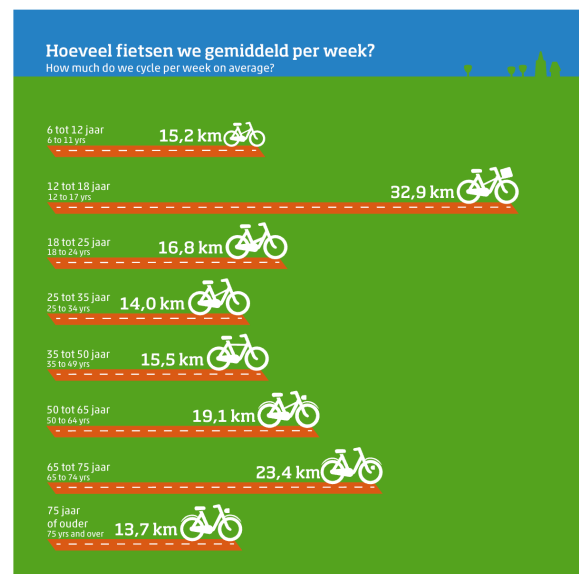
- | | |
|--|---|
| <ul style="list-style-type: none">• Filter unwanted data• Aggregate data (e.g., sum)• Group data based on a column• Sort rows based on a column• Concatenate data frames• Merge and join data frames• Quantize continuous values into bins | <ul style="list-style-type: none">• Scale column values• Resample time series data• Roll time series data in a window• Apply a transformation function• Use regular expressions• Drop rows or columns• Treat missing values |
|--|---|

Information visualization is a good way for both experts and laypeople to explore data and gain insights.



How much do we cycle per week on average?

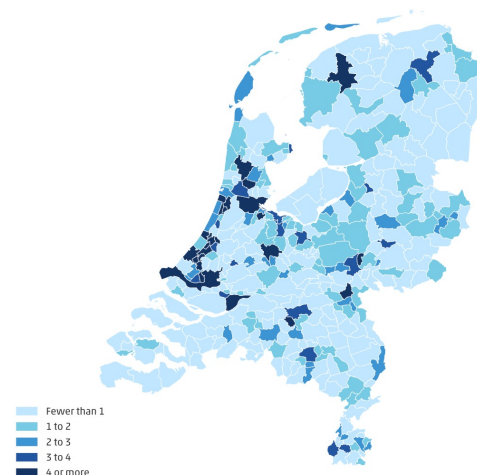
In 2020, the 12 to 17-year-olds cycled most frequently, either on bicycles or e-bikes. They also travelled the most kilometres: an average of 33 kilometres per person per week. The over-75s preferred the bicycle least. They cycled an average of 14 kilometres per week, as many as 25 to 34-year-olds.



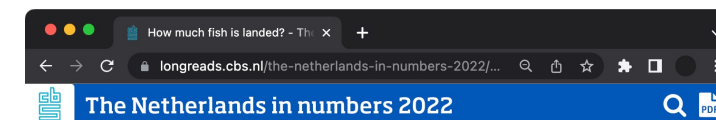
Average distance from a museum is 4 km

In 2020, Dutch people lived at an average distance¹ of 4.0 kilometres from a museum.² They could choose from an average of 3.5 Dutch museums within a travel distance of five kilometres. Noord-Holland had the most museums within this distance out of all the provinces, with an average of 8.9. Within this province, residents of Amsterdam enjoyed the most choice with 23.7 museums within five kilometres, followed by Haarlem residents with 6.1 museums. Zuid-Holland also offered relatively high choice with an average of 5.1 museums within five kilometres. In that province, the highest number of museums within this distance was seen in The Hague (13.4 museums) and in Leiden (9.7 museums).

Number of Dutch museums within a 5-km radius by road, 2020

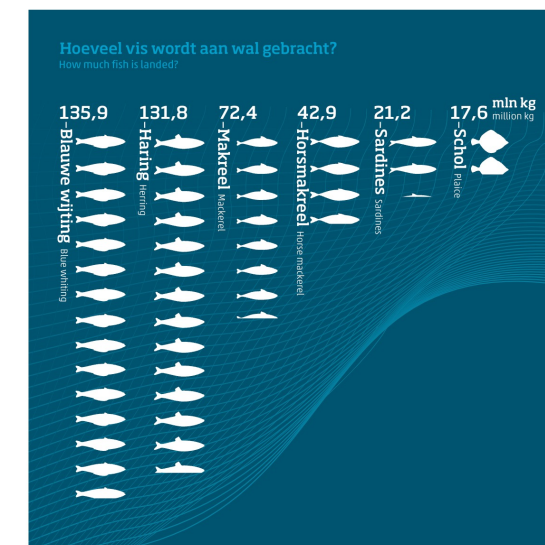


Show datatable



How much fish is landed?

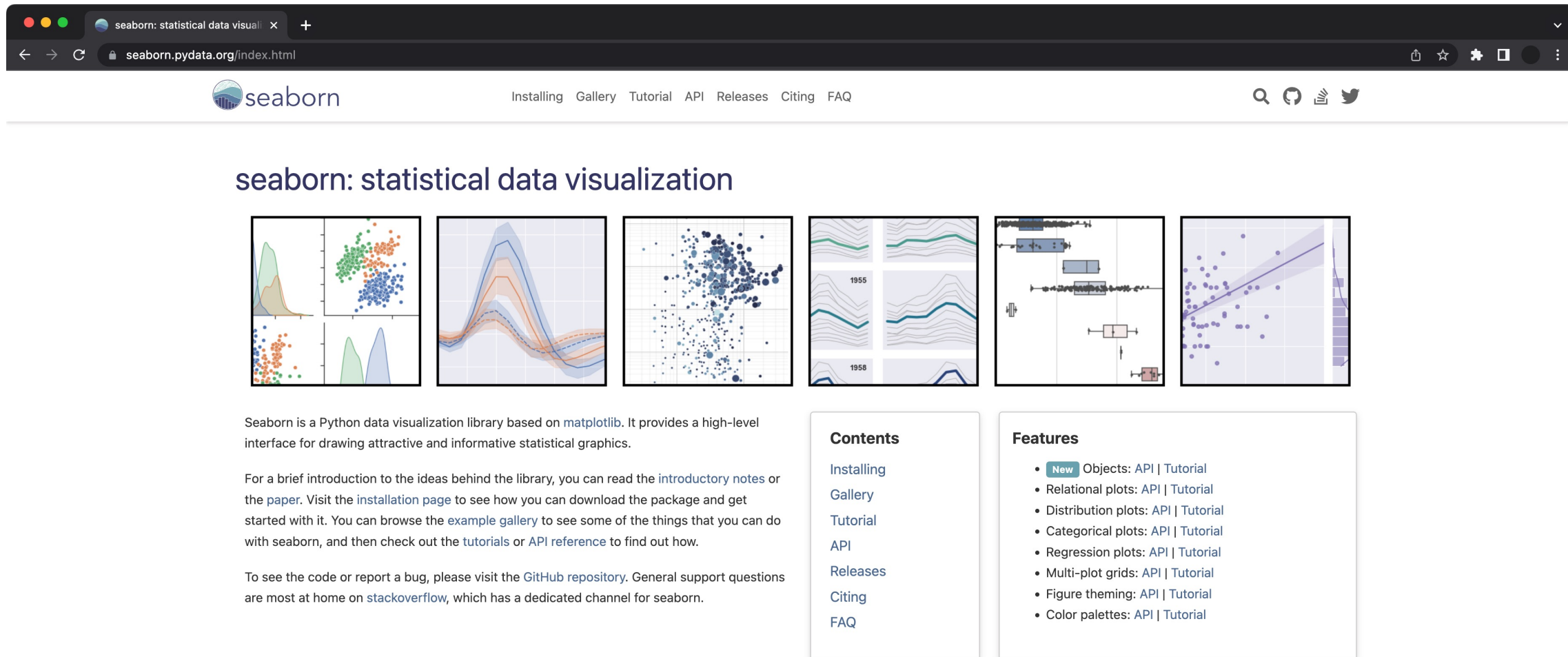
In 2021, the amount of fish brought into Dutch fishery ports from sea was 443.5 million kilograms. That is 17 percent more than in the previous year. Most of the fish, 91 percent, landed frozen via trawlers, with the rest landed fresh by cutters. Blue whiting was the most landed frozen fish at 135.9 million kilograms, plaice the most landed fresh fish at 17.6 million kilograms.



Trawlers landed 404.1 million kilograms of fish. These fishing vessels with funnel-shaped nets concentrate on schools of fish swimming in deeper waters. In 2021, they mainly landed blue

Explore Data

You can use the Python seaborn library (based on matplotlib) to quickly plot and explore structured data.



The screenshot shows the Seaborn website interface. At the top, there's a navigation bar with links: Installing, Gallery, Tutorial, API, Releases, Citing, and FAQ. Below this, the title "seaborn: statistical data visualization" is displayed. A row of six thumbnail images showcases different plot types: a joint plot with marginal histograms, a line plot with confidence intervals, a scatter plot with a regression line, a faceted line plot, a box plot, and a scatter plot with a regression line. Below the thumbnails, there's a paragraph describing Seaborn as a Python data visualization library based on matplotlib. To the right, there are two sections: "Contents" with links to Installing, Gallery, Tutorial, API, Releases, Citing, and FAQ; and "Features" with a list of capabilities like Objects, Relational plots, Distribution plots, Categorical plots, Regression plots, Multi-plot grids, Figure theming, and Color palettes.

seaborn: statistical data visualization

Seaborn is a Python data visualization library based on [matplotlib](#). It provides a high-level interface for drawing attractive and informative statistical graphics.

For a brief introduction to the ideas behind the library, you can read the [introductory notes](#) or the [paper](#). Visit the [installation page](#) to see how you can download the package and get started with it. You can browse the [example gallery](#) to see some of the things that you can do with seaborn, and then check out the [tutorials](#) or [API reference](#) to find out how.

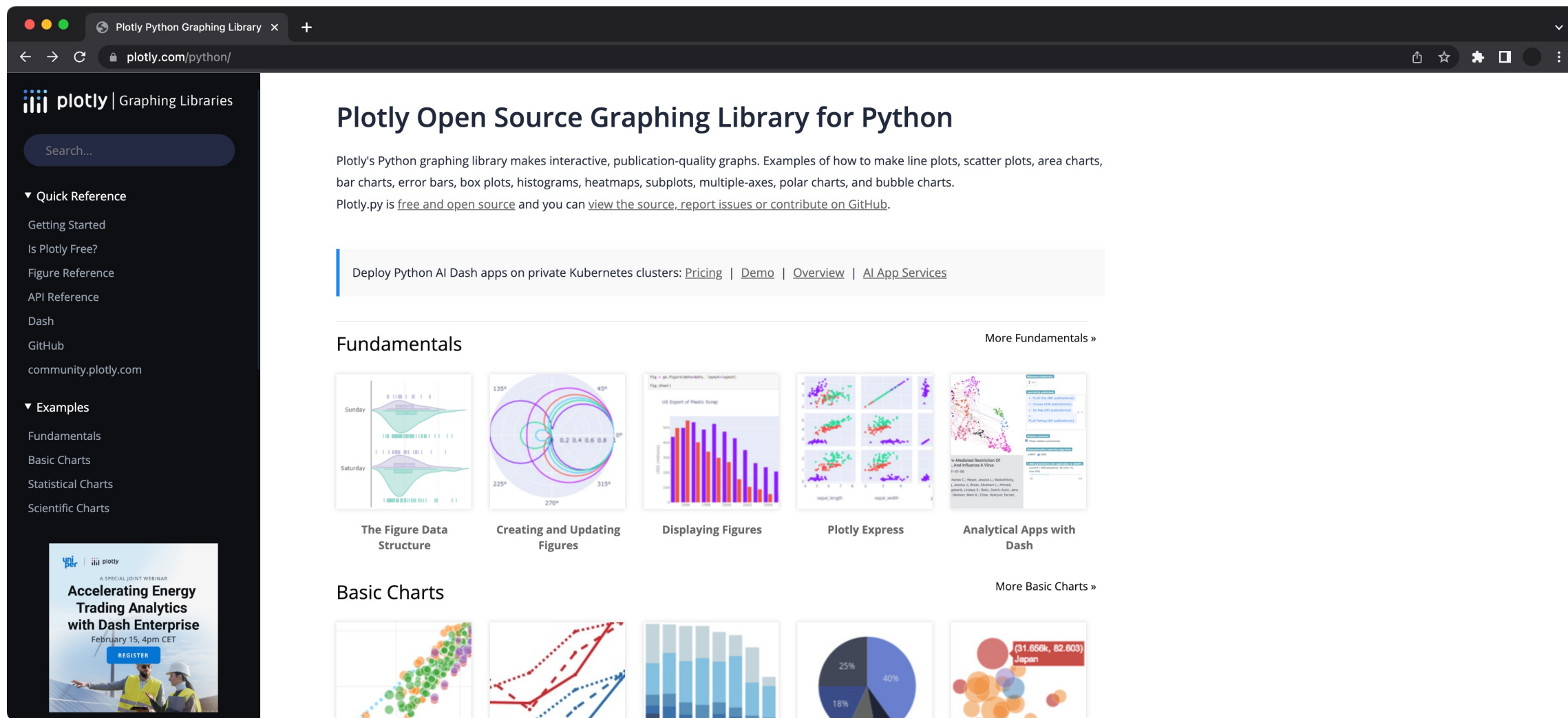
To see the code or report a bug, please visit the [GitHub repository](#). General support questions are most at home on [stackoverflow](#), which has a dedicated channel for seaborn.

Contents

- [Installing](#)
- [Gallery](#)
- [Tutorial](#)
- [API](#)
- [Releases](#)
- [Citing](#)
- [FAQ](#)

Features

- **New** Objects: [API](#) | [Tutorial](#)
- Relational plots: [API](#) | [Tutorial](#)
- Distribution plots: [API](#) | [Tutorial](#)
- Categorical plots: [API](#) | [Tutorial](#)
- Regression plots: [API](#) | [Tutorial](#)
- Multi-plot grids: [API](#) | [Tutorial](#)
- Figure theming: [API](#) | [Tutorial](#)
- Color palettes: [API](#) | [Tutorial](#)



The screenshot shows the Plotly Python Graphing Library website. The left sidebar contains a search bar and navigation links under 'Quick Reference' (Getting Started, Is Plotly Free?, Figure Reference, API Reference, Dash, GitHub, community.plotly.com) and 'Examples' (Fundamentals, Basic Charts, Statistical Charts, Scientific Charts). The main content area features the title 'Plotly Open Source Graphing Library for Python', a description of the library's capabilities, and links to deploy Python AI Dash apps. Below this are two sections: 'Fundamentals' and 'Basic Charts', each with five example visualizations and a 'More' link. The 'Fundamentals' examples include violin plots, a polar plot, a bar chart, a grid of scatter plots, and a network graph. The 'Basic Charts' examples include a scatter plot, a line plot, a bar chart, a pie chart, and a bubble chart.

Plotly Open Source Graphing Library for Python

Plotly's Python graphing library makes interactive, publication-quality graphs. Examples of how to make line plots, scatter plots, area charts, bar charts, error bars, box plots, histograms, heatmaps, subplots, multiple-axes, polar charts, and bubble charts. Plotly.py is [free and open source](#) and you can [view the source](#), [report issues](#) or [contribute on GitHub](#).

Deploy Python AI Dash apps on private Kubernetes clusters: [Pricing](#) | [Demo](#) | [Overview](#) | [AI App Services](#)

Fundamentals

More Fundamentals »

- The Figure Data Structure
- Creating and Updating Figures
- Displaying Figures
- Plotly Express
- Analytical Apps with Dash

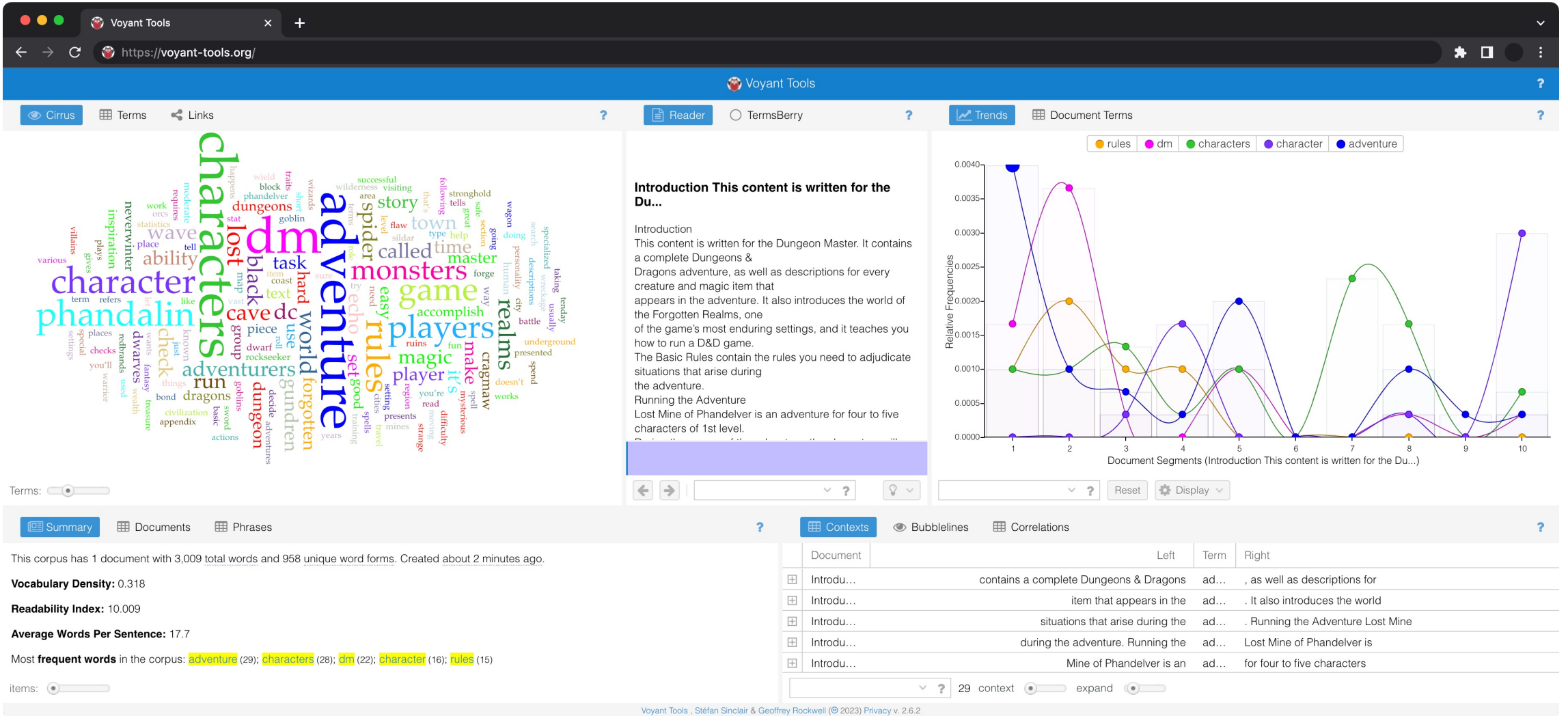
Basic Charts

More Basic Charts »

- Scatter Plot
- Line Plot
- Bar Chart
- Pie Chart
- Bubble Chart

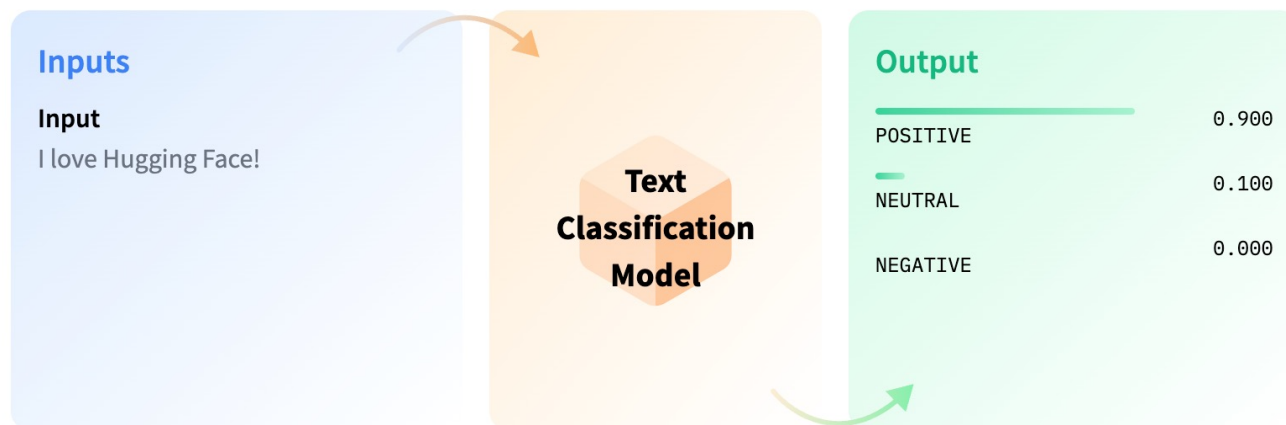
Explore Data

You can use the Voyant Tools to explore text data.

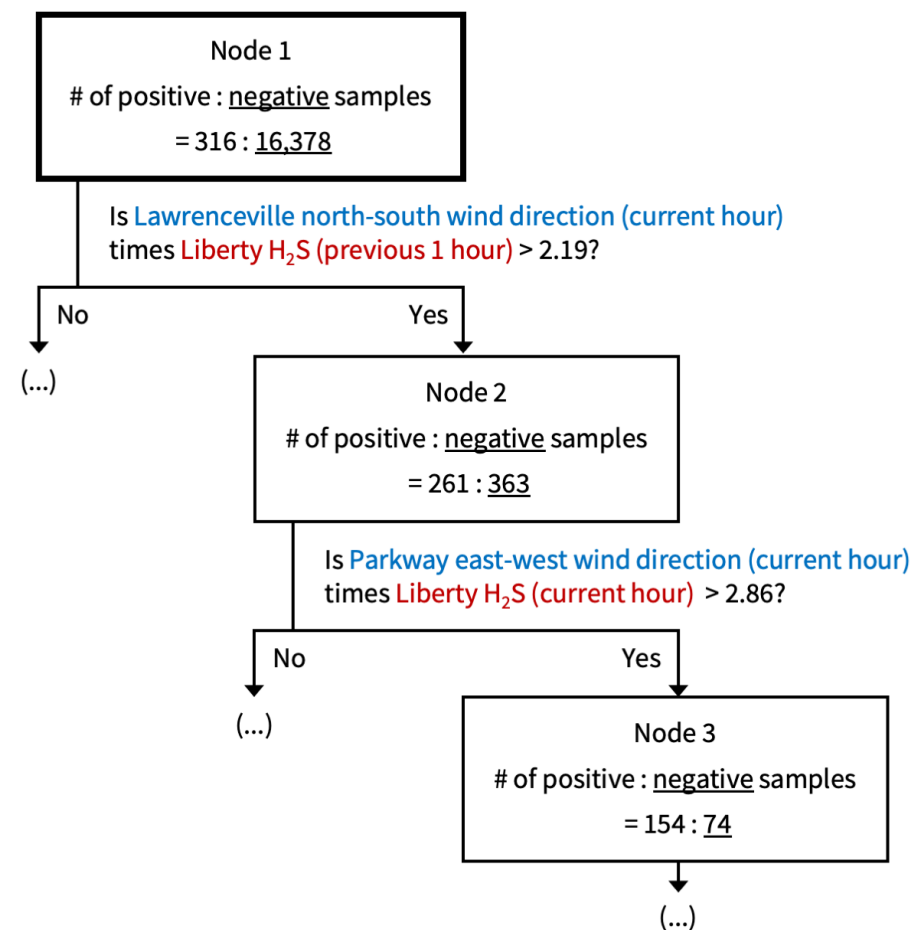


Model Data

This course will teach you machine learning techniques for modeling structured, text, and image data through three modules from a practical point of view.



Source -- <https://huggingface.co/tasks>



Source -- <https://smellpgh.org/>

Model Data

The structured data processing module uses air quality sensing and weather data to predict smell events.

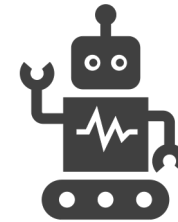
O ₃ : 26 ppb	CO: 127 ppb
H ₂ S: 0 ppb	PM _{2.5} : 9 µg/m ³
Wind: 17 deg	...

Observation 1

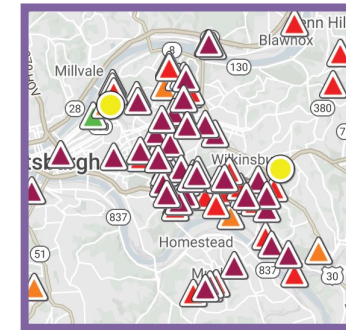
O ₃ : 1 ppb	CO: 1038 ppb
H ₂ S: 9 ppb	PM _{2.5} : 23 µg/m ³
Wind: 213 deg	...

Observation 2

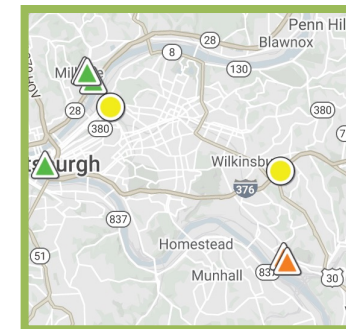
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Machine Learning



☹️ Has Event



😊 No Event

Model Data

The text data processing module is about **topic classification and**
online news items.

Class: Business



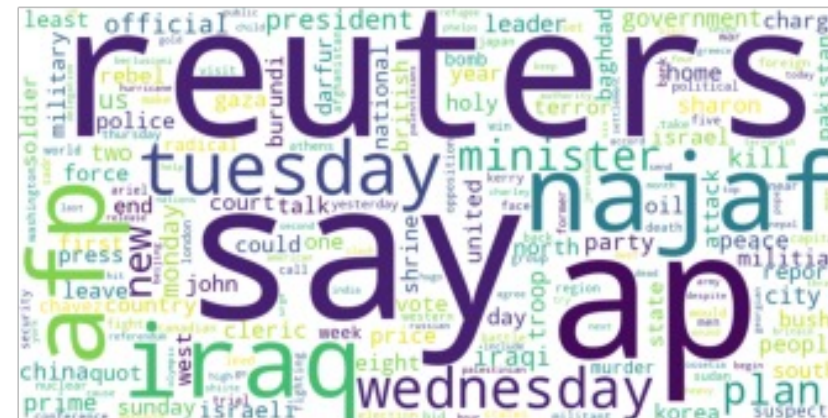
Class: Sci/Tec



Class: Sports



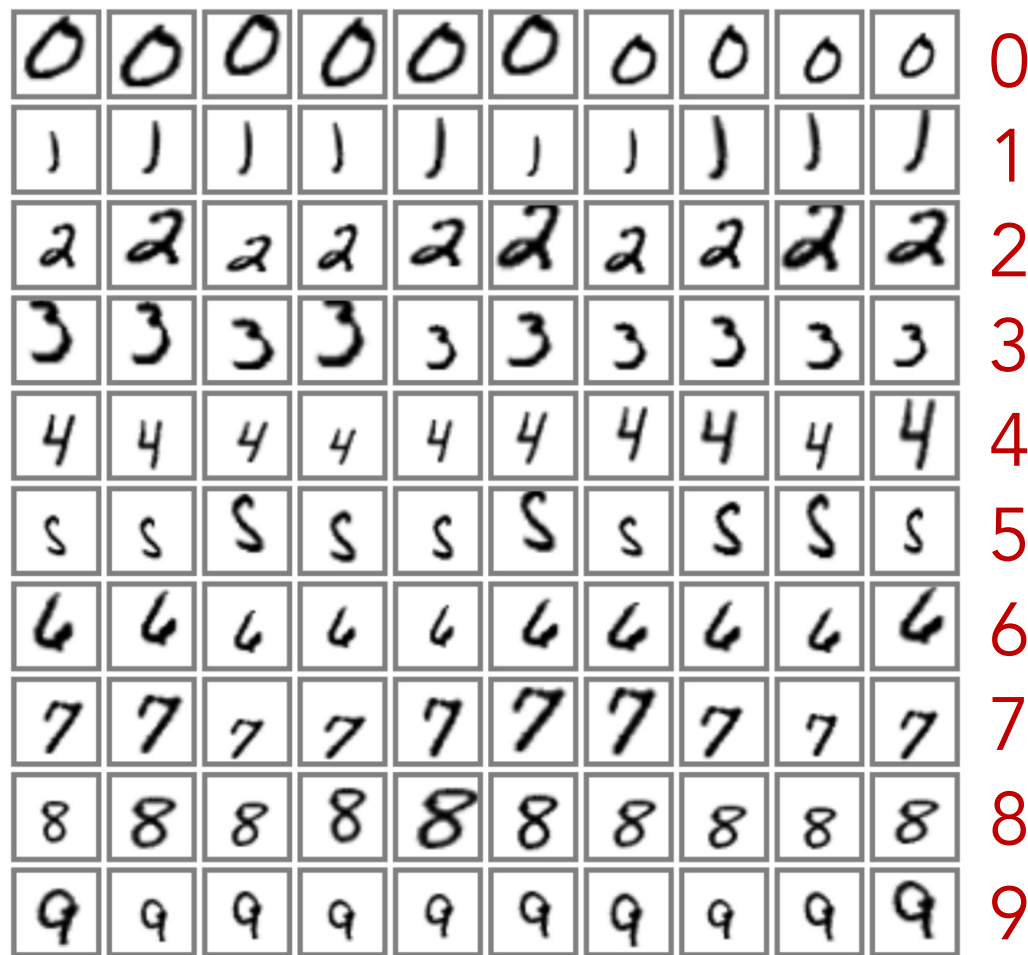
Class: World



Source -- <https://www.kaggle.com/datasets/amananandrai/ag-news-classification-dataset>

Model Data

The image data processing module uses deep neural networks to recognize digits from hand-written images.



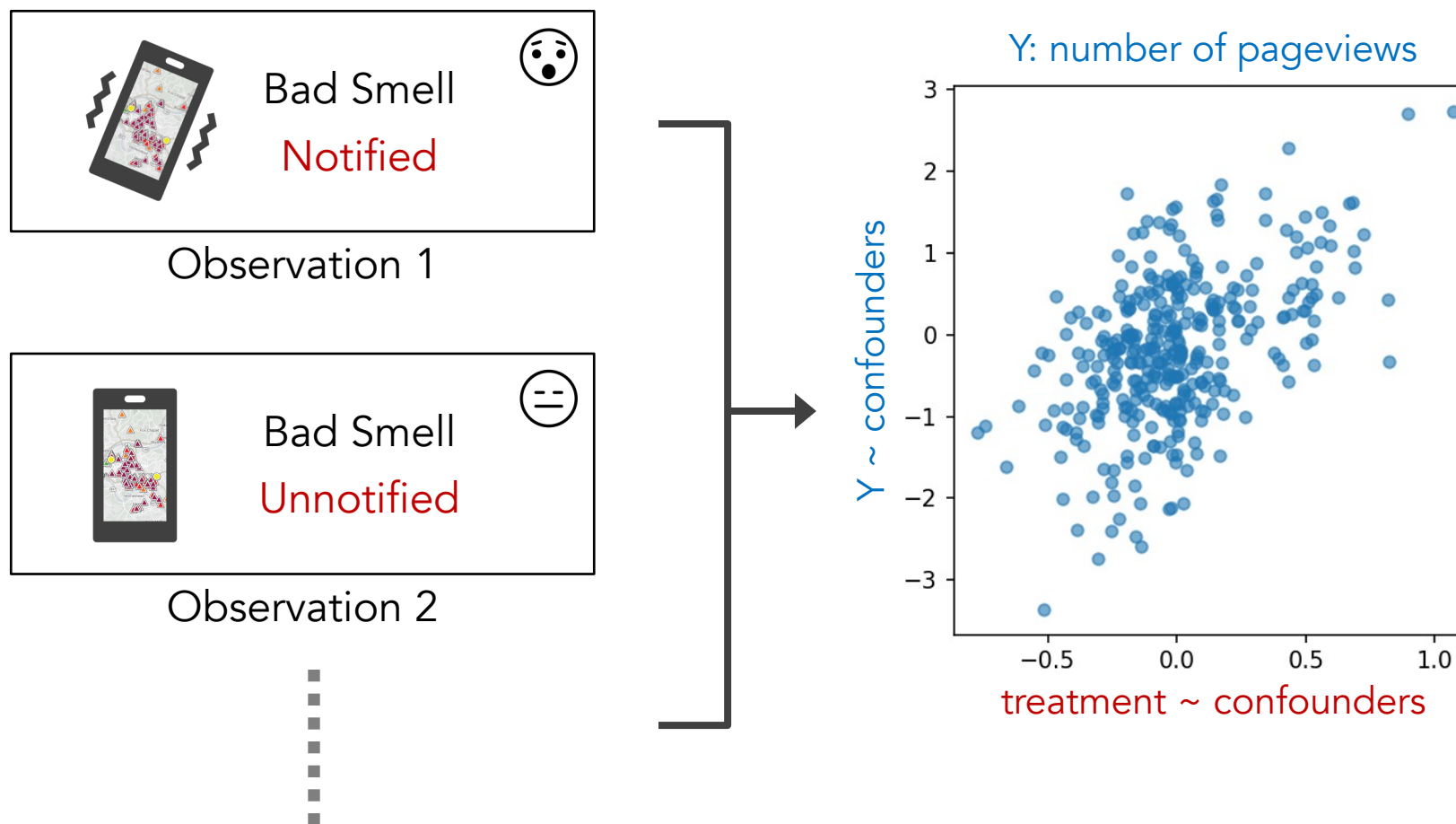
Model Data

A more complicated image data processing task is **fine-grained** and we use garbage classification as an example.



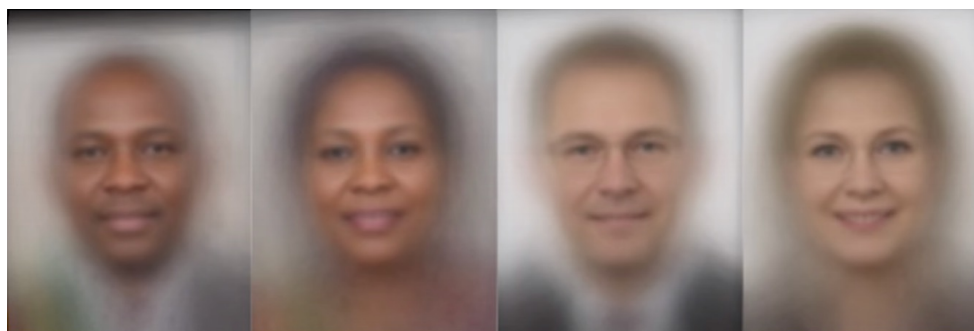
Deploy Models

Deploying models in the wild can enable further quantitative or qualitative research with insights, such as the push notification study in Smell Pittsburgh.



Deploy Models

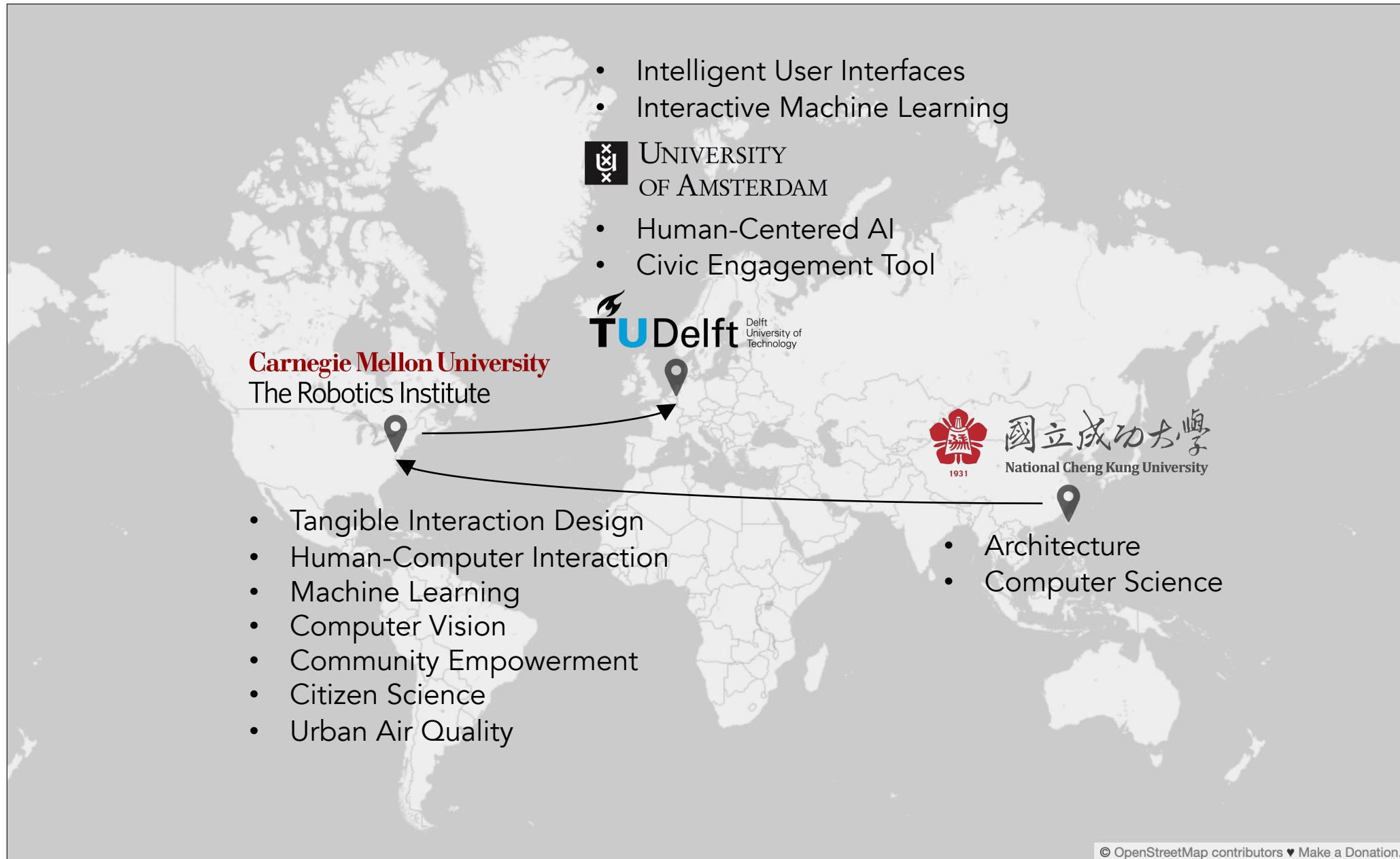
Another example is to study the behaviors of deployed systems to understand its social impact. For example, face recognition systems for recognizing gender did **worst on darker-skin female images**.



Gender Classifier	Darker Male	Darker Female	Lighter Male	Lighter Female	Largest Gap
 Microsoft	94.0% 	79.2% 	100% 	98.3% 	20.8% 
 FACE++	99.3% 	65.5% 	99.2% 	94.0% 	33.8% 
 IBM	88.0% 	65.3% 	99.7% 	92.9% 	34.4% 

This course has three teaching team members.
[Check Canvas for their contact information.](#)

- Yen-Chia Hsu, course coordinator
- TBA, teaching assistant
- TBA, teaching assistant



This course aims to familiarize you with various data science pipelines (processing structured data, text data, and image data) by alternating between theories and practices.

Schedule Outline

Week	Content	Deadline
1	Introduction + Data Science Fundamentals	
2	Structured Data Processing	
3	Deep Learning Overview + PyTorch	Reflective writing due, Tuesday 23:59
4	Mid-term Exam + Mid-term Exam Review (online)	
5	Text Data Processing	
6	Image Data Processing	Reflective writing due, Tuesday 23:59
7	Multimodal Data Processing + Guest Lecture	Reflective writing due, Tuesday 23:59
8	Final Exam	

Administration

- Announcements will be made on Canvas.
- Lectures will be streamed and recorded with links on the Canvas home page (quality not guaranteed).
- Lectures may be given virtually if unexpected situations happen, same as seminars.
- Use TicketVise (preferred) or email to ask questions.
- There is no attendance requirement.
- You are expected to treat others with mutual respect and appreciation regardless of any differences.
- It is strongly recommended to stay home if you have symptoms associated with respiratory infections.

Grading

- Assessment includes **two exams** (mid-term 40%, final 50%) and **three reflective writing submissions** of assignments (10% total).
- According to OER (Onderwijs- en Examenregeling), **you need to get at least 5.5 in BOTH the practical and theory parts to pass the course**. This means:
 - Theory part: the weighted sum of the scores of the mid-term and final exams must be at least 5.5
 - Practical part: the weighted sum of the scores of all reflective writing submissions for the assignments must be at least 5.5

Exams

- Check <https://multix.io/data-science-book-uva/#schedule-outline> about the coverage of exams.
- Exams are based on multiple choice questions. We provide mock exams for you to practice.
- You may bring an A4-size cheat sheet with two sides of content (written or printed) to the exams. No other materials are allowed during the exams (except the cheat sheet). Also, no calculators.
- Check the date and time of the exams carefully on UvA DataNose.
- No minimum grade requirement for each exam to pass the course. There is a resit, which is 90% weight (will override your original weighted sum of exam scores).
- We will use guess correction for the exams (check the syllabus for details).
- You will have access to Jupyter Notebook during the exam.

Exam Coverage Range

Lecture	Topic	Mid-term	Final	Resit
1	Introduction	yes	yes	yes
2	Data Science Fundamentals	yes	yes	yes
3	Structured Data Processing (Part I)	yes	yes	yes
4	Structured Data Processing (Part II)	yes	yes	yes
5	Deep Learning Overview	yes	yes	yes
6	PyTorch Basics	yes	yes	yes
7	Text Data Processing (Part I)	no	yes	yes
8	Text Data Processing (Part II)	no	yes	yes
9	Image Data Processing (Part I)	no	yes	yes
10	Image Data Processing (Part II)	no	yes	yes
11	Multimodal Data Processing	no	yes	yes

Reflective Writing Submissions for Assignments

- We only grade your reflective writing submissions (pass/fail) but not the assignments.
- Use the reflective writing template that we provide for submissions.
- You need to show that you have done the assignment by explaining what you have learned.
- We only accept submissions on Canvas (no email submissions). Always double-check your submissions by downloading them and checking if they work.
- Check the syllabus for the late submission policy (automatic deduction of 10% max points per day).
- Assignment materials can appear in the mid-term exam, final exam, and resit.
- Share your GenAI usage experiences for this course in the reflective writing.

Important Notes on Expectations

- We do not teach programming. Instead, we teach how to do data science using programming and machine learning techniques. Basic concepts of machine learning will be covered.
- We do not aim to cover everything in data science. Instead, we introduce ways of doing data science that enable students to study further in relevant topics.
- We do not teach you data collection. Instead, we assume that someone has collected the datasets.
- We do not teach data science in production. Most of the techniques that are covered in this course are for the development environment.
- We expect students to have good Python programming skills already. The Python Coding Warm-Up practice reflects our expectations.

GenAI Usage and Policy

- We allow students to use GenAI tools, such as ChatGPT, Gemini, Claude, UvA AI Chat, etc.
- GenAI tools can hallucinate information, generate wrong content, or provide fake sources/citations. You are responsible for fact-checking and verifying all AI-generated content.
- The official lecture slides and course readings remain the single source of truth. If the AI generates content that contradicts the course materials, you must follow the course materials.
- The teaching team is neither responsible nor liable for your exam errors resulting from reliance on unverified AI-generated content.

Math, Coding, and Course Readings in the Exam

- If a math equation is on the slide, and we explained it in the lecture, it means that the equation may appear in the exam (in different ways). You also need to know what they are doing at a high level.
- Multiple-choice questions in the exam can contain coding-related questions. Only the code in the “Task Answers” pages for all modules will be covered in the exam (check the syllabus for details).
- This course has many readings. The exams will be based on the lecture slides (primarily) and the required course readings (secondly). Optional course readings are used for GenAI fact-checking.
- If a concept appears on the lecture slides, it may appear on the exam, and the course readings are used to support the concept with more explanations and details.
- If a concept appears in the course readings but not on the slides, it will not appear on the exam.

For more details, check the course website and the syllabus page below:

- <https://multix.io/data-science-book-uva/>
- <https://multix.io/data-science-book-uva/syllabus.html>



Questions?